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Use l'Hôpital's Rule to find the limit.

$$\lim_{x \rightarrow 3} \frac{2x - 6}{9x^2 - 81}$$

L'Hôpital's Rule states that if $f(a) = g(a) = 0$, f and g are differentiable on an open interval I containing a , and $g'(x) \neq 0$ on I if $x \neq a$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}, \text{ assuming that the limit on the right side of this equation exists.}$$

First, check that $\lim_{x \rightarrow 3} \frac{2x - 6}{9x^2 - 81}$ is an indeterminate form. Since $\lim_{x \rightarrow 3} \frac{2x - 6}{9x^2 - 81}$ is in the form $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)}$, identify $f(x)$ and $g(x)$.

$$f(x) = 2x - 6$$

$$g(x) = 9x^2 - 81$$

Next check that $f(a) = g(a) = 0$ by substituting 3 into $f(x)$ and $g(x)$.

$$2x - 6 = 9x^2 - 81$$

$$2(3) - 6 = 9(3)^2 - 81$$

$$0 = 0$$

Since $f(a) = g(a) = 0$ is true, then $\lim_{x \rightarrow 3} \frac{2x - 6}{9x^2 - 81}$ is an indeterminate form and L'Hôpital's Rule can be applied. Find $f'(x)$ and $g'(x)$. Differentiate f .

$$f(x) = 2x - 6$$

$$f'(x) = 2$$

Now differentiate g .

$$g(x) = 9x^2 - 81$$

$$g'(x) = 18x$$

$$\text{Thus, } \lim_{x \rightarrow 3} \frac{2x - 6}{9x^2 - 81} = \lim_{x \rightarrow 3} \frac{2}{18x} = \frac{1}{27}.$$